computing the biased distribution $P_{\rm GM}(x)$ of the generative model enables unbiasing x_i in a last post-processing step [163]. This ensures that we are precisely sampling from $P(x)$.

In practice the method is only efficient if the weights are almost uniformly distributed. This requires that the generative distribution is as close as possible to the target distribution. To this aim, the machine is trained to minimize the Kullback-Leibler (KL) divergence between the generative and target distributions. Because the KL divergence between two distributions is not symmetric, two choices can be made: (i) Maximum likelihood training, in which one minimizes $D_{KL}(P||P_{\rm GM})$. This approach has the advantage that $P_{\rm GM}$ has to cover well all the support of $P(x)$. However, it also requires existing samples from $P(x)$, which renders training impractical: we want to train a machine to sample from the target, but for this we need to be able to sample from the target. (ii) Variational or energy-based training, where one minimizes $D_{KL}(P_{\rm GM}||P)$, which corresponds to the free energy of the generative model. While this choice does not require sampling from $P_{\rm GM}$ it has the important drawback that the generative model might only cover part of the support of the target, which is known as mode collapse.

To deal with these problems, several architectures and training strategies have been proposed. To sample a two-state protein model Noé *et al.* [163] focused on a normalizing flow architecture to represent the map $f(z)$, and a training strategy based on mixing the variational approach with maximum likelihood, for which experimental structures and short molecular dynamics simulations were used. In subsequent work, the group introduced equivariant flows to implement physical symmetries [165], and used a higher-temperature Boltzmann distribution as a prior [166, 167] (see also Refs. [168, 169]). To sample crystalline structures, it was recently proposed to generate displacements from a reference lattice structure instead of absolute particle positions [170]. Gabrié *et al.* [171] introduced a more efficient training strategy based on mixing standard Monte Carlo moves with moves proposed by the generative model. Applying Boltzmann generators to sample supercooled liquids yields performances in the same magnitude as previously known enhanced sampling techniques [172].

Several studies have focused on other models in which sampling is challenging, e.g. spin-glasses, hard optimization problems and lattice field theories [154, 164, 173–185]. It remains unclear how these methods compare to standard ones, or whether the efficiency is universal or model dependent [186–189].

Finally, it could be worth combining the ML models discussed in Sec. III, which can precisely predict future dynamics, with generative models. The former can be used to detect active regions or even particles which are likely to rearrange, while the latter can subsequently propose new configurations based on local rearrangements.

DATA AVAILABILITY

The dataset "GlassBench" and python scripts used to create the benchmarks presented in Sec. V is publicly available and can be download on zenodo (https://doi.org/10.5281/zenodo.10118191).

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